

Hybrid grey wolf and whale optimization for enhanced Parkinson's prediction based on machine learning models using biomedical sound

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ARTICLE INFO

Keywords:

Feature selection
Biomedical voice measurements
Machine learning model
Parkinson's disease
Whale optimization
Grey wolf optimization
Prediction
Hybrid optimization algorithm

ABSTRACT

Background: Biomedical voice measurements have been used by many physicians and scientists to distinguish Parkinson's patients from ordinary people. Measurements of biomedical voices involve many variables calculated from signal analysis of the voice. These variables can be used to distinguish Parkinson's patients from non-Parkinson's patients. Unfortunately, using all computed variables may be ineffective and inaccurate due to the complexity of establishing a relationship between all the input variables and Parkinson's states.

Methods: This paper describes the development of a hybrid optimizer by combining two optimization algorithms: the Grey Wolf Optimizer and the Whale Optimizer. The hybrid optimizer enhances feature selection to provide a fast and robust Parkinson's prediction model. Additionally, this research incorporates five other feature selection algorithms for comparison purposes: Ranker, Greedy, BestFirst, Hybrid Grey Wolf Optimization, and Whale Optimization. The outputs from these algorithms are fed into six prediction models to determine the most accurate combination. These models include the neural network, Quest, Chi-squared Automatic Interaction Detection, support vector machine, CR-tree, and logistic regression models. Subsequently, the developed models are compared to identify the most accurate model based on various performance metrics.

Results: The combination of the hybrid grey wolf and whale models yielded the highest scores in most metrics, achieving a perfect recall and a high F1 score. All models generated similar output, with an accuracy greater than 0.89. Quest, CR-tree, and neural networks are the most reliable and accurate models.

Conclusions: Biomedical sound measurements can be used to develop an accurate and cost-effective Parkinson's prediction model.

1. Introduction

The human brain is the central control unit of the body. Therefore, if any of its components are slightly damaged, it will profoundly impact all other organs of the body. Parkinson's disease (PD) is a progressive and chronic neurological condition that explicitly targets dopamine receptors (Alzubaidi et al., 2021 [1]; Loh et al., 2021 [2]).

Statistics of Parkinson's disease diagnoses show that it affects approximately 0.3 % of the world's population and 1 % of people above 60 years (around 6 million people). These statistics rank PD as the second worst neurodegenerative disease after Alzheimer's (Chintalapudi et al., 2022 [3]; Aarsland et al., 2021 [4]; Tolosa et al., 2021 [5]; Tysnes and Storstein 2017 [6]; Das, 2010 [7]; Sadjia, 2006 [31], Suganya and Sumathi [32]).

The symptoms of Parkinson's disease (PD) progressively worsen over time and can be categorized into two types based on whether it affects non-motor functions or motor functions. The motor symptoms of PD are associated with the depletion of dopamine in the striatum and the degeneration of the substantia nigra. On the other hand, non-motor symptoms of PD encompass orthostatic hypotension, constipation, urinary dysfunction, hyposmia, memory loss, sleep disturbances, pain, anxiety, and depression (Chintalapudi et al., 2022 [3]; Tolosa et al., 2021 [5]). Moreover, cognitive impairment could arise and affect one's daily living activities, learning capacity, judgment, memory, language, thinking, and behavior, leading to emotional difficulties (Kurmi et al., 2022 [8]; Aarsland et al., 2021 [4]; Kalia and Lang 2015 [9]).

Although the main causes of Parkinson's disease (PD) are generally unknown, recent studies have identified factors related to mutually

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<https://doi.org/10.1016/j.imu.2024.101524>

Received 21 March 2024; Received in revised form 18 May 2024; Accepted 19 May 2024

Available online 20 May 2024

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genetic and environmental (Sadek et al., 2019 [10]; Kalia, 2015 [9]). In addition, due to the absence of a definitive medical test for accurately diagnosing the early stages of Parkinson's disease (PD), misclassification errors in diagnosis are relatively infrequent, with error rates between 15 % and 24 % (Chintalapudi et al., 2022 [3]; Tolosa et al., 2021 [5]; Rizzo et al., 2016 [11]).

Accordingly, attempts to overcome the previously mentioned difficulties have been made through the considerably growing attention among healthcare services to applying artificial intelligence to enhance the accuracy of predictions of neurological disorders such as PD. Early diagnosis helps combat the disease by relieving symptoms and mitigating complications while allowing patients and their families to deal with the new situation more properly (Alzubaidi et al., 2021 [1]; Loh et al., 2021 [2]; Sadek et al., 2019 [10]).

This study aimed to create a predictive model that can anticipate the occurrence of Parkinson's disease by using voice analysis, machine learning, and feature selection models. The contributions of this paper are.

- The development of a new optimizer that integrates two existing optimization models, specifically, whale optimization [36] and hybrid Grey Wolf Optimization [37], to enhance the efficiency of the previous optimizer.
- The identification of the most significant features from voice analysis variables through combinations selected by six optimization models.
- The construction of machine learning prediction models, such as neural network, support vector machine, Quest, CHAID, CR-tree, and logistic regression, to analyze and predict outcomes effectively.

This paper is organized as follows. Section 2 provides the literature review. Section 3 describes the feature selection algorithms, the proposed hybrid optimizer, and the prediction models used in the study. This section also introduces the performance evaluation metrics. Section 4 details the performance of the optimization and prediction models and presents the main findings. Finally, Section 5 presents the conclusions.

2. Literature review

Medical reports have disclosed that the number of Parkinson's patients is expected to continually increase, especially in developed countries, and the medical treatment cost is anticipated to accelerate as a consequence. Hence, all interested parties are trying to find a way to improve early diagnoses of this disease. In the last two decades, researchers have relied heavily on machine learning algorithms to make it easier to differentiate between the symptoms of Parkinson's disease and other neurological diseases such as tremors. Such an achievement could improve public health through the accurate diagnosis of various diseases. Numerous researchers have explored various diseases by applying machine learning models. Specifically, COVID-19 has been a focal point of investigations, as evidenced by studies conducted by Al-Najjar and Al-Rousan (2020) [12], Al-Najjar et al. (2020) [13], Al-Najjar et al. (2021) [14], and Al-Rousan et al. (2020) [15].

ÇAĞLAR et al. (2010) [16] aimed to discriminate against Parkinson's patients depending on the biomedical voice of humans. Two types were used, namely a multilayer perceptron network and a radial basis function network. The utilization of linguistic hedges in the adaptive neuro-fuzzy classifier enables its effectiveness in feature selection from the dataset and in achieving remarkable recognition outcomes. Employing the adaptive neuro-fuzzy classifier with linguistic hedges yielded an exceptional classification performance, with training and test data accuracy rates of 95.38 % and 94.72 %, respectively. In addition, Bocklet et al. (2011) [17] proposed a Parkinson's disease prediction method that uses a support vector machine (SVM) and correlation-based classification. The best outcomes provided an accuracy rate of 90.5 %.

Ramani et al. (2012) [18] emphasized the significance of different feature selection and classification algorithms when working with

Parkinson's monitoring data. Algorithms, such as s-Nearest Neighbors algorithm (KNN), random tree, SVM, and partial least squares regression, were utilized to predict biomedical voice measurements. Filtering techniques were implemented on these algorithms to enhance their classification capabilities. The primary outcomes supported that random tree was the best at predicting Parkinson's disease, yielding a 100 % accurate classification. Rustempasic and Can (2013) [19] introduced an automated machine-learning approach that accurately identified Parkinson's disease based on the speech or voice of patients. They proposed fuzzy C-means clustering and a pattern recognition approach for detection. The results of combining the two methods were promising, achieving 68.04 % accuracy.

Tarigoppula et al. (2013) utilized various machine-learning approaches to diagnose Parkinson's disease (PD) by analyzing biomedical voice measurements. The findings indicated that the support vector machine (SVM) exhibited a favorable accuracy rate of 88.9 %, surpassing the majority and k-NN algorithms. Additionally, the random forest classification algorithm demonstrated a commendable accuracy of 90.26 %, while Naïve Bayes exhibited the lowest accuracy of 69.23 %. Sharma et al. (2014) applied three classifiers based on Multiple Pattern recognition, SVM, and K Nearest Neighbor to develop a precise Parkinson's detection based on biomedical voice signals. The findings indicate a significant level of accuracy (85.294 %).

Tiwari, A. (2016) [20] developed a prediction model based on feature selection algorithms. Hence, multiple machine learning models were applied, including SVM, MLP, boosting, bagging, rotation forest, random forest, random subspace, and decision tree. Based on the main observations, the optimal approach was the random forest algorithm with 20 selected features, which achieved an accuracy rate of 90.3 %. Furthermore, Mathew's correlation coefficient and ROC values were calculated as 0.73 and 0.96, respectively. Meanwhile, Cai Z. et al. (2018) [21] enhanced many artificial intelligence methods, including the fuzzy k-nearest neighbor method and the evolutionary instance-based learning approach based on vocal measurements to diagnose Parkinson's cases. The simulation results revealed that the proposed approach performed better than other models.

Sadek R. et al. (2019) [10] presented a Parkinson's predictor that employed an artificial neural network system utilizing the back-propagation algorithm. The dataset, comprising 195 samples, was split into 170 train samples and 25 test samples. The proposed design of the neural network system significantly improved the system's robustness. Sonu et al. (2017) [22] aimed to use data mining techniques to predict PD cases. The conclusions supported using decision tree and building a classifier, resulting in 88–94 % accuracy. They sought to utilize data mining techniques, including KNN, logistic regression, decision tree, SVM, and naive Bayes, to predict whether an individual is healthy or diagnosed with Parkinson's. Their works employed data mining techniques to predict whether individuals are healthy or afflicted with Parkinson's disease. The conclusions supported using a decision tree and building a classifier (88–94 % accuracy).

Ayap et al. (2021) [23] highlighted using multilayer perceptron (MLP) on biomedical human voice markers diagnosed with PD. Their main aim was to reduce the number of loss cases to be undiagnosed properly. The results showed that the average accuracy was 91.5 % when the proposed prediction model was used. Chintalapudi et al. (2022) [3] improved the accuracy of disease predictions by applying machine learning (ML) algorithms. The researchers created three cascaded neural network models for deep learning. These models were trained using selected datasets containing voice biomarkers of Parkinson's disease (PD) patients to assess diagnostic accuracy. The results indicated that the LSTM model outperformed the other models, achieving an accuracy of 99 %. Kurmi et al. (2022) [8] introduced deep learning (DL) models for Parkinson's disease prediction by utilizing DaTSCAN images. The outputs from the initial stage were then passed through a fuzzy fusion logic-based approach, which improved overall classification performance, with an accuracy of 98.45 %.

The authors of [41] explored Parkinson’s disease (PD), a chronic neurological disorder affecting movement, characterized by rigidity, bradykinesia, loss of balance, and resting tremors. Diagnosing PD is challenging due to intricate brain patterns, and the authors achieved only a 25 % error rate. The study proposed a convolutional neural network based on a computer-aided diagnostic method utilizing T2-weighted MRI slices from the Parkinson’s Progression Markers Initiative. A freehand region of interest technique was used to select the midbrain region. An evaluation using standard metrics like accuracy, sensitivity, specificity, and area under the curve revealed the superior performance of the Convolution neural network (CNN) compared to traditional machine learning methods.

Martinez-Eguiluz et al. [42] evaluated machine learning algorithms’ abilities to predict Parkinson’s disease (PD) based on non-motor symptoms. They analyzed nine algorithms with varying granularity levels using data from Biocrates (96 subjects) and PPMI (687 subjects). Most algorithms demonstrated excellent accuracy, with multilayer perceptron and support vector machine achieving the best results (86.3 % and 84.7 %, respectively). Feature selection improved performance and reduced variables. The researchers combined Biocrates and PPMI data to moderately enhance algorithm performance, particularly in recall. The study identified crucial discriminative features for potential screening tools and underscored the efficacy of machine learning in reliably predicting Parkinson’s disease (PD) using non-motor measures.

Meanwhile, Brien et al. [43] explored biomarkers for PD, PD with mild cognitive impairment, and dementia (PDD). The classifier achieved an ROC-AUC of 0.88 after analyzing data from 106 healthy controls and 121 PD patients, with higher AUC values for cognitive dysfunction subgroups. It demonstrated 83 % sensitivity and 78 % specificity, with a high AUC of 0.95 for PDD. The classifier’s confidence scores predicted both motor and cognitive function in PD. The study suggested that a non-invasive eye-tracking task could serve as an additional screening method for PD and related cognitive impairment.

The literature has emphasized the significant impact of machine learning on the fundamental aspects of medical sciences and research in identifying Parkinson’s patients. Nevertheless, many researchers have reported that 70–90 % of patients showed an affected voice, which supports that voice and prosody are the earliest symptoms of Parkinson’s disease. Therefore, this research applies various machine-learning methods based on the biomedical voice of humans to predict Parkinson’s disease (PD).

Tuncer et al. [44] devised an innovative approach for Parkinson’s disease (PD) detection using vowels. Their method combined two hybrid models to extract salient features: minimum average maximum tree and singular value decomposition (SVD). These extracted features were then fed into the K-nearest neighborhood algorithm to construct a prediction model. The model underwent tenfold validation during training to enhance its classification capabilities. The highest prediction rates were 92.46 % (without post-processing) and 96.83 % (with post-processing). Loh et al. [45] developed a Parkinson’s disease prediction model using a two-dimensional convolutional neural network and EEG signals. The dataset they collected contains 31 cases, 15 of which were PD patients. The researchers used a tenfold validation model to enhance the PD training model. The results showed a high classification accuracy of 99.46 % and revealed that the model can accurately predict PD based on the patient’s status.

Kaplan et al. [46] developed a Parkinson’s classifier using a nested patch division to improve the feature selection model and develop a high-accuracy classifier. They used three datasets from MRI images to detect PD symptoms. Two classification algorithms, namely k-nearest neighbor (KNN) and support vector machine (SVM), were used with the extracted feature to develop the prediction models. The results showed that the highest accuracy was around 98 %.

The above literature review underscores the rising prevalence of Parkinson’s disease, particularly in industrialized nations, which has increased medical expenses. Over the past two decades, researchers

have predominantly employed machine learning algorithms to distinguish Parkinson’s symptoms from the symptoms of other neurological illnesses. Various studies have utilized models like support vector machine and multilayer perceptron to achieve accuracy levels ranging from 69.23 to 100 %. The use of feature selection and classification methods, such as random trees, to improve prediction has been emphasized. Recent advancements, including deep learning and convolutional neural networks, demonstrate accuracy levels of up to 99 % and 98 %, respectively.

The consensus among studies is that machine learning is significant in early Parkinson’s detection, particularly when it is used to analyze speech and voice patterns. Furthermore, the scientific community consistently highlights the potential of these approaches. The current study aims to predict Parkinson’s disease using diverse machine-learning approaches based on biomedical voice data. Previous research focused on using optimization models to improve the performance of prediction models using fewer features. Table 1 summarizes the previous models used in the literature mentioned throughout this section.

All previous models attempted to develop prediction models using machine learning without selecting the optimal variables to enhance the quality of predictions with fewer features. However, the developed models exhibited very high complexity due to the large number of features, and the feature-selection process needs to be grounded in scientific concepts. Moreover, researchers across various disciplines have demonstrated that the same machine learning models can yield excellent results when the appropriate features are selected when developing prediction models, as evidenced in Refs. [47–49].

This study aims to fill the gaps in the previous models by developing a hybrid optimizer based on the grey wolf optimizer and whale optimizer to enhance Parkinson’s prediction. In doing so, this study aims to enhance the accuracy of Parkinson’s prediction and extend the implications of predictive analytics beyond the healthcare domain.

Table 1
Literature review summary.

Author	Year	Objectives	Algorithm
Chintalapudi et al. [3]	2022	Predict PD	RNN, MLP, and LSTM
Kurmi, A. et al. [8]	2022	Predict PD	ResNet50, Inception-V3, Xception, VGG16, and fuzzy fusion logic
Ayap et al. [23]	2021	Predict and diagnose PD	Multilayer perceptron neural network (MLP)
Sonu et al., 2017 [22]	2020	Predict PD	naïve Bayes, logistic regression, decision tree, support vector machine, and KNN.
Sadek R. et al. [10]	2019	Identify PD	Neural network
Cai Z. et al. [21]	2018	Diagnose PD	Different machine learning (ML) and optimization models
Tiwari, A [20].	2016	Prediction of PD	Different ML models
Sharma et al. [24]	2014	Detection of Parkinson	Different ML models
Rustempasic & Can [19]	2013	Detect the PD	Fuzzy C-means clustering and pattern recognition
Ramani et al. [18]	2011	classifying the severity of PD	SVM, KNN, random tree, partial least square regression
Bocklet et al. [17]	2011	Detection of PD	Support vector machine (SVM) and correlation-based classification
ÇAĞLAR et al. [16]	2010	Detection of PD	Different ML
Hussain et al. [41]	2023	Detection of PD	Neural network
Martinez-Eguiluz et al. [42]	2023	Detection of PD	Different machine learning models
Brien et al. [43]	2023	Classification and staging of PD	Machine learning models with eye-tracking

3. Research methodology

This section describes the development of a Parkinson's disease prediction model that utilizes feature extraction characteristics associated with general voice disorders observed in Parkinson's disease patients. This chapter is divided into four subsections: dataset collection and analysis, developing Parkinson's disease prediction model, the proposed PD model, and performance evaluation.

3.1. Dataset collection and analysis

The PD dataset is collected from Little et al., 2008 [25]. The dataset contains a range of biomedical voice measurements from 31 people, 23 of whom are denoted as PD. Various people's recordings were recorded. A total of 195 audio recordings were obtained and divided into 147 recordings of PD cases and 48 recordings of healthy individuals. The variables for the main descriptive statistics of the PD dataset are shown in Table 2. In addition, a Pearson correlation was used to understand the relationship between the input variables and the status of the participants. The results show that the correlation between status and input variables is between -0.4 and 0.6 for the studied variables. The results also show that there are weak relationships with the status. Therefore, the linear relationship is not useful in determining relationships between status and input variables. Moreover, the correlations between independent variables range from weak to strong, which creates a multi-collinearity problem. This motivated the researchers to develop a nonlinear model to find the most important variables that can affect predictions of Parkinson's cases.

3.2. Parkinson's disease prediction based on the attribute evaluator and search method

The biomedical voice measurements of the Parkinson's dataset contain 21 independent variables and one dependent variable (status). Since using all variables to develop a prediction model increases the prediction model's complexity, it was necessary to select variables from among the independent variables that highly control the dependent variables. Therefore, the researchers adopted feature selection to select highly correlated variables with dependent variables to create an efficient and less costly model for predicting Parkinson's cases.

Thus, this study applied feature selection methods by selecting the most important attributes. The feature selection process was divided

into two phases: the attribute evaluator and search method phases. The attribute evaluator aims to assess the relationship between the studied variables, and the search method aims to investigate different independent variable combinations to find the best combinations to use. Based on our previous analysis, the process of creating a linear regression Parkinson predictor using a simple linear regression model is inefficient. Therefore, we adopted different nonlinear methods to predict Parkinson's cases by using an attribute evaluator and search method. We used different prediction models to predict Parkinson's cases, including the QUEST model, support vector machine, logistic regression, CHAID, neural network, and CR-tree. Moreover, the attribute evaluator uses a classifier attribute evaluator and subset evaluator (Hall, 1998 [26]), by which two search methods are used, mainly the BestFirst search and greedy stepwise search. The BestFirst search method aims to search the attribute subsets using greedy hill climbing augmented with a backtracking model, whereas the greedy stepwise method aims to search backward or forward through the studied attributes.

The proposed evaluator and search methods were chosen after testing different combinations between different models in the previous work, as discussed by Chetty et al. (2015) [27] and Onik et al. (2015) [28]. Moreover, this study developed three models, as shown in Table 3. Each model ranks and ranges the selected independent attributes from the most effective to least effective variable based on the relationship with Parkinson's cases. After the proposed models were developed, as shown in Table 2, all the prediction models were used with these models to determine the best prediction model based on the feature selection methodology. All the collected results were compared using different performance metrics to find the most important variables based on the developed model (i.e., Ranker, BestFirst, greedy, Whale optimization, and Hybrid GWO). The results were compared to find the most important predictors (independent variables) in developing the prediction model.

3.3. The Proposed Optimization Model

Feature selection was enhanced by combining two models to improve the performance of predicting Parkinson's disease. The proposed model comprises two phases: the selection phase and the refine phase. In the selection phase, the Whale Optimization Algorithm is employed to generate 30 solutions and forward the best solution (determined by the accuracy of the prediction model) to the refining phase. Subsequently, the best solution is fed into the hybrid GWO model to regenerate another 30 solutions based on the output of the Whale Optimization Algorithm. The best solution, as determined by the hybrid GWO, is then selected as the optimal model for predicting Parkinson's disease. The two phases are intended to enhance the selection process by identifying the best feature combinations through a two-level selection approach, as shown in Fig. 1(a).

The dataset was initially partitioned into two segments to construct the prediction model: the training and test data. The training data were used to train the prediction models with past information, while the test data were employed to validate the performance of the developed models with future input. We used 70 % and 30 % of the dataset to create the training and test data, respectively. The study flowchart is illustrated in Fig. 1(b).

When GWO and whale optimization algorithms were developed to

Table 2
Statistics of the collected dataset.

	Min	Max	Mean	Std. Dev.
MDVPFoHz	88.333	260.105	154.229	41.390
MDVPFhiHz	102.145	592.030	197.105	91.492
MDVPFloHz	65.476	239.170	116.325	43.521
MDVPJitter	0.002	0.033	0.006	0.005
MDVPRAP	0.001	0.021	0.003	0.003
MDVPPPQ	0.001	0.020	0.003	0.003
JitterDDP	0.002	0.064	0.010	0.009
MDVPSshimmer	0.010	0.119	0.030	0.019
NHR	0.001	0.315	0.025	0.040
HNR	8.441	33.047	21.886	4.426
ShimmerAPQ5	0.006	0.079	0.018	0.012
MDVPAPQ	0.007	0.138	0.024	0.017
ShimmerDDA	0.014	0.169	0.047	0.030
spread1	-7.965	-2.434	-5.684	1.090
DFA	0.574	0.825	0.718	0.055
MDVPSshimmerdB	0.085	1.302	0.282	0.195
ShimmerAPQ3	0.005	0.056	0.016	0.010
D2	1.423	3.671	2.382	0.383
PPE	0.045	0.527	0.207	0.090
Status	0	1	0.75	0.432
spread2	0.006	0.450	0.227	0.083
RPDE	0.257	0.685	0.499	0.104

Table 3
Feature selection methods.

Models	Search method	Attribute Selection
Ranker [34,35]	Ranker	ClassifierAttributeEval
BestFirst [34,35]	BestFirst	Subset evaluator
Greedy [34,35]	Greedy stepwise	Subset evaluator
Whale Optimization Algorithm [36]	–	–
Hybrid GWO [37]	–	–

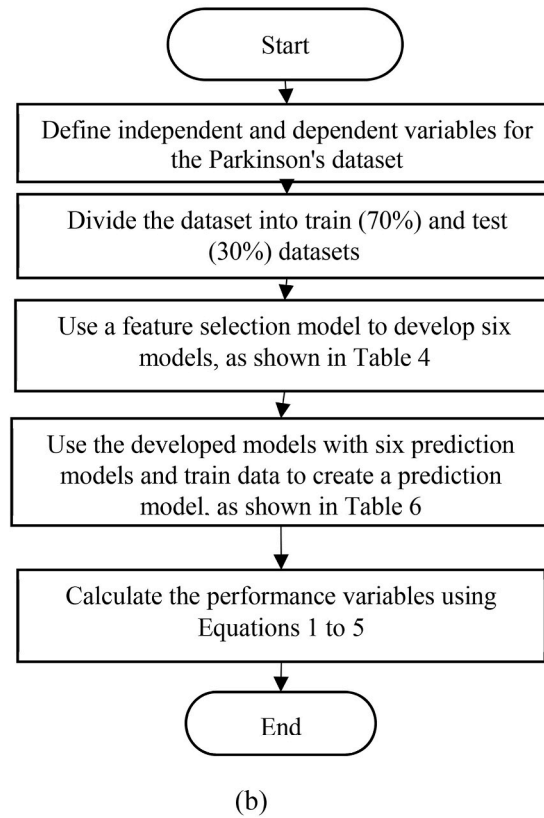
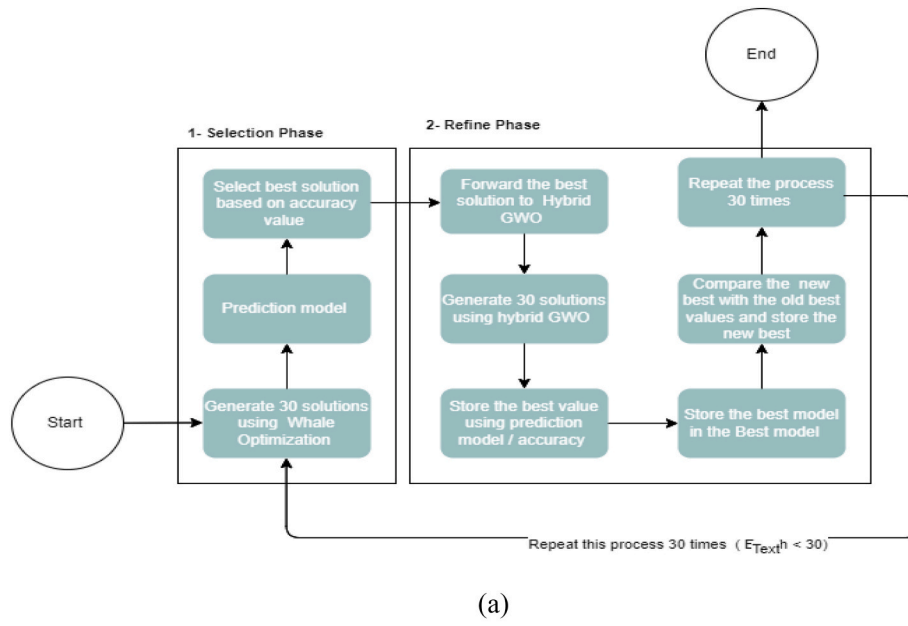


Fig. 1. The method used to build (a) the proposed hybrid optimization and (b) the methodology to build a Parkinson's prediction model.

enhance Parkinson's disease prediction, the algorithms were set and initial parameters were tuned to achieve the best results for the proposed model. The initial values for the whale optimization algorithm's search agents, function, and iterations were 30, F1, and 500, respectively. The initial values for GWO were 30, F10, and 500, respectively. For more information about the code and initial values for these optimization algorithms, please see the following link, which explains the main functions and necessary settings: <https://seyedaslimirjalili.com/>. The complexity of the two algorithms was calculated by adding the complexity of the whale optimization with grey wolf optimization.

Therefore, the complexity of the proposed algorithm was calculated as shown in Eq. (1). The number of maximum iterations was selected independently for each algorithm.

$$\begin{aligned} \text{Complexity} = & O(\#whales * \#dimension * \text{maximum number of iteration}) \\ & + O(\#wolves * \#dimension * \text{maximum number of iteration}) \end{aligned} \quad (1)$$

Finally, the proposed model aims to integrate two optimization algorithms (Grey wolf optimization and Whale optimization) to improve Parkinson's disease prediction. The model is expected to be used for

medical applications, specifically in diagnosing Parkinson's disease. Two main features were used to evaluate the prediction models' performance: performance metrics comparison and variable importance analysis. The performance metrics mentioned in Section 3.4 below were also used to evaluate the performance of the models.

3.4. Performance metric

The performance of PD prediction can be assessed using various performance metrics and constructing a confusion matrix. The confusion matrix illustrates the correspondence between correct and incorrect classification outcomes. It comprises four components: false positive (FP), true positive (TP), true negative (TN), and false negative (FN). This study used only specific classification parameters, including recall, precision, accuracy, and F1 score [29,30,33]:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (2)$$

$$precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$False Omission rate (FOR) = \frac{TN}{TN + FN} \quad (5)$$

$$F1 \text{ score} = 2 * \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

Where TP is the number of cases correctly predicted as Parkinson's disease, TN is the number of cases correctly predicted as not having Parkinson's disease, FP is the number of non-Parkinson's cases predicted as Parkinson's cases, and FN is the number of non-Parkinson's cases predicted as Parkinson's cases.

4. Results analysis and discussion

This section analyzes the developed models based on the feature selection methodology. Feature selection is used to create models containing different numbers of predictors using different attribute evaluators and search methods. The results of the first step are used with different prediction models to find the best classification model that needs the fewest predictors and the least complex architecture.

4.1. Results of the developed models

The first step in developing the proposed model was to run the feature selection models based on the suggested attribute evaluators and the search methods, as shown in Table 3. The results show that the Ranker model suggests using all the independent variables. The Ranker method ranked the independent jobs based on importance as follows (from most to least important): PPE, Jitter-DDP, MDVP-Shimmer, Shimmer-APQ3, Shimmer-APQ5, MDVP-RAP, MDVP- PPQ, MDVP: Jitter(%), MDVP: Flo(Hz), MDVP: Fhi(Hz), MDVP:Shimmer(dB), D2, spread1, spread2, MDVP:APQ, DFA, RPDE, HNR, NHR, Shimmer:DDA, and MDVP:Fo(Hz).

In addition, the results from the BestFirst model show that only three independent variables—DFA, SPEED 1, and SPEED 2—were selected to build prediction models. In contrast, the Greedy model shows that only five independent variables were to be used: DFA, SPEED 1, SPEED 2, MDVP:RAP, and SHIMMER: APQ3. Finally, after the used models were compared, the results revealed only three independent variables in all the developed models: DFA, SPEED 1, and SPEED 2 (Table 4). Each prediction model used the forwarded variables to predict Parkinson's cases, as presented in Table 5. Tables 4 and 5 show the most important

Table 4

The generated models based on the feature selection.

Models	Variables	Number of variables
Ranker [34,35]	All variables	21
BestFirst [34,35]	DFA, SPEED 1, SPEED 2	3
Greedy [34,35]	DFA, SPEED 1, SPEED 2	5
Whale optimization [36]	MDVP:RAP, SHIMMER:APQ3	15
	D2, DFA, MDVP:APQ, MDVP:Jitter(Abs), MDVP:PPQ, MDVP:RAP, spread1, MDVP:Shimmer, NHR, RPDE, Shimmer:APQ3, Shimmer:APQ5, Shimmer:DDA, MDVP:Jitter(%), spread2	
Hybrid GWO [37]	MDVP:Flo(Hz), Jitter:DDP, MDVP:Jitter (Abs), MDVP:Shimmer, Shimmer:APQ3, MDVP:APQ, Shimmer:DDA, DFA spread1, spread2, MDVP:Shimmer(dB), D2, MDVP:PPQ	13
Proposed model (Whale-Hybrid GWO)	Jitter:DDP, MDVP:Fhi(Hz), MDVP:Fo (Hz), MDVP:PPQ, MDVP:RAP, MDVP: Shimmer(dB), Shimmer:APQ5, spread1, Shimmer:APQ3, Shimmer:DDA, NHR	11

Table 5

The independent variables using all optimization models based on different prediction models.

Prediction Model	Important design metrics	Design information.
SVM	Kernel-Polynomial	Epsilon: 0.05, gamma, 0.05 Regularization parameter: 5 Stop criteria: 1×10^{-2}
Neural network	Two hidden layers	Neural network mode: multilayer perceptron
Quest	Single tree	Max and min branch: 2 and 1
CHAID		Categorical combining: voting
CR-tree		Continuous combining: mean
Logistic regression	Binomial	Scale: deviance, Tolerance 1×10^{-9} , Binomial procedure: enter method

and capable variables for building prediction models.

4.2. The Parkinson's prediction model based on the Ranker model

The developed model used a Ranker feature selection associated with Parkinson's disease. After all the incoming variables were ranked and all unranked variables were eliminated, the variables were forwarded to prediction models. The main prediction models used were SVM, neural network, Quest, CHAID, CR-tree_status, and logistic regression. Only 70 % of the data were used to develop a prediction model. The results of the prediction models show that SVM, neural network, and logistic regression used all the input variables, whereas the Quest, CR-tree, and Chi-squared automatic interaction detection (CHAID) models used part of the input variables. Subsequently, the test dataset was employed to assess the proficiency of the prediction models in accurately predicting Parkinson's disease cases.

The train dataset results show that all prediction models' precision, accuracy, recall, FOR, and F1 scores ranged between 0.77 and 0.99, as

Table 6

The results of the training dataset based on the Ranker model.

	Accuracy	FOR	Recall	F1_score	Precision
SVM	0.88	0.90	0.98	0.92	0.87
Neural network	0.90	0.85	0.96	0.94	0.91
Quest	0.86	0.77	0.94	0.92	0.89
CHAID	0.92	0.84	0.95	0.94	0.95
CR-tree	0.93	0.96	0.99	0.96	0.92
Logistic_Reg	0.90	0.83	0.95	0.94	0.92

shown in Table 6. The results show that all the developed models can predict Parkinson's cases. Furthermore, the CR model exhibited superior performance across accuracy (0.93), precision (0.92), recall (0.99), FOR (0.96), and F1 score (0.92) metrics.

The test dataset showed similar behavior, as the performance metrics of all prediction models were between 0.67 and 0.98, as shown in Table 7. The results show that the trained prediction models can be used to predict Parkinson's cases. The Quest model achieved the best performance. The training and testing datasets confirm that all the models can be used to predict Parkinson's disease, though the highest performance was achieved by the Quest model since it showed more stability and reliability between training and test results.

The analysis of important variables in Parkinson's disease cases reveals that the developed models yielded varying sets of significant variables, as illustrated in Fig. 2. The six most important variables for SVM, in descending order of importance, were Low_spread1, PPE, MDVP: Fo(Hz), MDVP: Flo(Hz), DFA, and MDVP: APQ. In the case of the neural network, the most important variables were spread1, PPE, spread2, MDVP: APQ, D2, and Jitter: DDP. For CHAID the most important variables were MDVP: Fhi(Hz), PPE, DFA, MDVP: Flo(Hz), spread2, and NHR. For the Quest model, the most noteworthy variables were spread1, PPE, MDVP: PPQ, MDVP: Jitter(%), NHR, and spread2. For the CR_Tree model, spread2, MDVP: Flo(Hz), MDVP: Fo(Hz), D2, DFA, and Jitter: DDP were the most important variables. Finally, For the logistic model, Jitter: DDP, MDVP: Shimmer Shimmer: DDA, spread1, PPE, and spread2 were important for developing a Parkinson's model.

The results show that spread1, PPE, and spread2 are the most influential variables across all models. The analysis shows that each model has different important variables, and no model shows that the same variables have the same level of significance. Therefore, the prediction model with the highest performance metrics is the best, and its variables are among the most important. The results indicate that the Parkinson's prediction model developed using the Quest model was the best. Moreover, it was found that nine variables are sufficient to build a prediction model, as presented in Table 5.

4.3. The Parkinson's prediction model based on the BestFirst model

The BestFirst model was used to develop a predictive model for PD and showed the most important variables of Parkinson's cases. The results show that only three important variables are directly related to Parkinson's cases: speed1, speed2, and DFA (Table 4). The output of the BestFirst model was forwarded to five prediction models to develop a Parkinson's prediction model. The results of the train and test datasets are shown in Tables 8 and 9. The train dataset results show that the accuracy, precision, recall, FOR, and F1 scores of all prediction models ranged between 0.76 and 0.99, as shown in Table 8. The results also show that all the developed models can predict Parkinson's cases. The CR-tree model produced the best performance, with accuracy, recall, precision, FOR, and F1 score values of 0.93, 0.99, 0.92, 0.96, and 0.96, respectively (Table 8).

The test dataset showed similar behavior, as the accuracy, precision, recall, FOR, and F1 scores of all prediction models ranged between 0.67 and 0.98, as shown in Table 9. The results show that the trained prediction models can predict Parkinson's cases. The Quest model achieved

the highest performance, as its accuracy, precision, recall, and FOR values were 0.91 and 0.90, 0.98, 0.92, and 0.92, respectively.

The importance variables analysis of the PD prediction models showed that all the models returned different values (Fig. 3). However, all models showed that speed 1, speed 2, and DFA are important variables. The results of the train dataset, test dataset, and important variables confirm that all the models can be used to predict PD, though the best performance was achieved by using the CR model, which showed more stability and reliability than the other models. In contrast, the Quest model did not show any important variables, indicating that this model is built without creating tree rules and using one node without depending on the input variables. Therefore, the Quest model does not help classify Parkinson's cases based on the variables used in this study.

4.4. The Parkinson's prediction model based on the greedy model

The greedy model was used to develop a predictive model for Parkinson's disease. The results show that only five important variables are directly related to Parkinson's cases: speed1, speed2, DFA, MDVP: RAP, and SHIMMER: APQ3 (Table 4). The train results are shown in Tables 9 and 10. The train dataset results show that all prediction models' accuracy, precision, recall, FOR, and F1 scores ranged between 0.5 and 1 (Table 10). The results show that all the developed models can predict Parkinson's cases except the Quest model. The CR-tree model yielded the best performance, as its accuracy, recall, precision, FOR, and F1 score values were 0.91, 0.93, 0.95, 0.83, and 0.94, respectively (Table 10).

The test dataset showed similar behavior, as all the studied variables ranged between 0.50 and 1, as presented in Table 11. The results show that the trained prediction models can be used to predict Parkinson's cases. The highest performance metrics were achieved using SVM, CR-tree, and logistic regression, as the accuracy, recall, precision, and FOR results were 0.89, 0.96, 0.90, 0.85, and 0.92, respectively. The test dataset showed that all three models (SVM, CR-tree, and logistic regression) exhibited similar performance results.

The analysis of the importance of the variables is shown in Fig. 4. The Quest model was eliminated from the analysis since the tree model did not use input variables to create a prediction model. Meanwhile, the rest of the models showed different results. NN, logistic regression, and SVM used five input variables to develop prediction models, whereas the CHAID and CR_tree models used three input variables. The results of the train dataset, test dataset, and the important variables confirm that all the models can be used to predict PD. The best performance was achieved using the CR-tree model, which used only three variables, namely MDVP: RAP, speed 1, and SHIMMER: APQ3. This model also outperformed the others in terms of stability and reliability.

4.5. The Parkinson's prediction model based on the whale optimization model [36]

The Whale model was used to develop a predictive model for Parkinson's disease. This model showed that 15 important variables are directly related to Parkinson's cases, as shown in Table 4. The training results are shown in Table 12. The train dataset results show that all prediction models' accuracy, precision, recall, FOR, and F1 scores ranged between 0.9 and 1, as shown in Table 12. The results show that all the developed models can predict Parkinson's cases, except the Quest model. The CR-tree model performed the best, with accuracy, recall, precision, FOR, and F1 score values of 0.91, 0.95, 0.93, 0.83, and 0.94, respectively, as presented in Table 12.

The test dataset showed similar behavior, as all the studied variables ranged between 0.70 and 1, as presented in Table 13. The results show that the trained prediction models can be used to predict Parkinson's cases. The best performance metrics are achieved using CHAID and CR-tree, as the results of accuracy, precision, recall, FOR, and F1 score were 1. The test dataset showed that neural network, CHAID, and SVM have

Table 7
The results using the test dataset based on the Ranker model.

	Accuracy	FOR	Recall	F1_score	Precision
SVM	0.88	0.90	0.98	0.92	0.87
Neural network	0.88	0.75	0.92	0.92	0.92
Quest	0.91	0.92	0.98	0.94	0.90
CHAID	0.89	0.85	0.96	0.92	0.90
CR-tree	0.89	0.76	0.92	0.92	0.94
Logistic_Reg	0.84	0.67	0.88	0.9	0.91

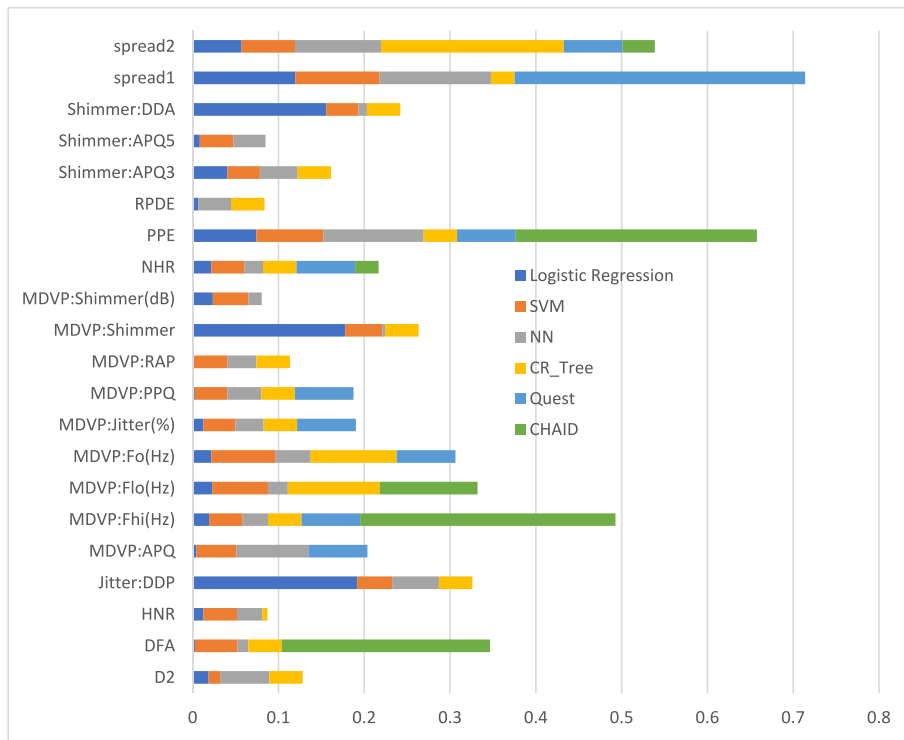


Fig. 2. Importance variables based on the Ranker model.

Table 8

The results of the developed models using the training dataset based on the BestFirst model.

	Accuracy	FOR	Recall	F1_score	Precision
SVM	0.88	0.90	0.98	0.92	0.87
Neural network	0.90	0.85	0.96	0.94	0.91
Quest	0.86	0.77	0.94	0.92	0.89
CHAID	0.92	0.84	0.95	0.94	0.95
CR-tree	0.93	0.96	0.99	0.96	0.92
Logistic_Reg	0.90	0.83	0.95	0.94	0.92

Table 9

The results of the developed models using a test dataset based on the BestFirst model.

	Accuracy	FOR	Recall	F1 score	Precision
SVM	0.88	0.90	0.98	0.92	0.87
Neural network	0.88	0.75	0.92	0.92	0.92
Quest	0.91	0.92	0.98	0.94	0.90
CHAID	0.89	0.85	0.96	0.92	0.90
CR-tree	0.89	0.76	0.92	0.92	0.94
Logistic_Reg	0.84	0.67	0.88	0.9	0.91

similar performance results, while optimal values were achieved using CR-tree.

Fig. 5 shows the importance of the variables analyzed in the PD prediction model. Speed 1 was the most important variable for all the models, and the least important variables were DFA, RPDE, and MDVP: Shimmer.

4.6. The Parkinson's prediction model based on the Hybrid Grey Wolf Optimization Model [37]

The hybrid GWO model was used to develop a predictive model for Parkinson's disease. The Hybrid GWO results show that 13 important variables were directly related to Parkinson's cases, as shown in Table 4.

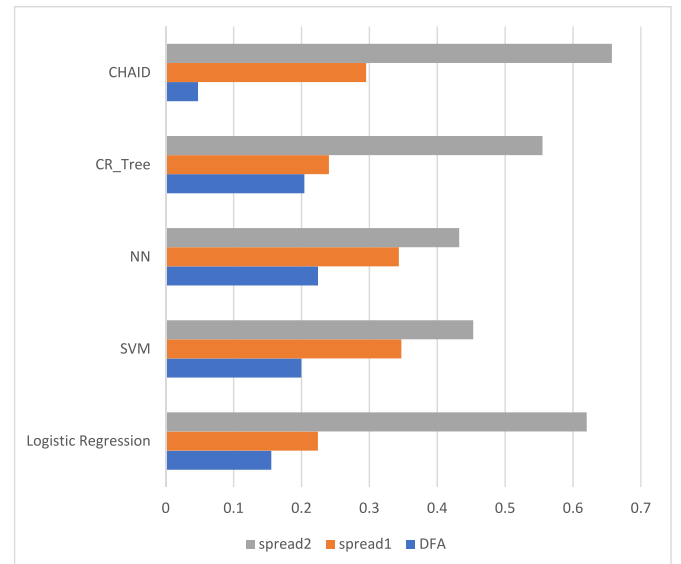


Fig. 3. Importance variables based on BestFirst Model.

Table 10

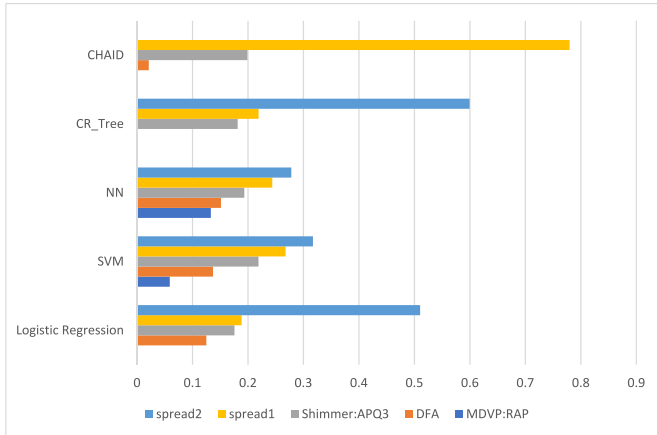
The results of the developed models using a training dataset based on the Greedy model.

	Accuracy	FOR	Recall	F1_score	Precision
SVM	0.85	0.78	0.95	0.90	0.87
Neural network	0.89	0.76	0.92	0.92	0.94
Quest	0.76	NaN	1.00	0.86	0.76
CHAID	0.89	0.85	0.96	0.94	0.90
CR-tree	0.91	0.83	0.95	0.94	0.93
Logistic_Reg	0.84	0.70	0.92	0.90	0.88

Table 11

The results of the developed models using a testing dataset based on the greedy model.

	Accuracy	Recall	FOR	F1_score	Precision
SVM	0.89	0.96	0.85	0.92	0.90
Neural network	0.83	0.88	0.65	0.88	0.89
Quest	0.75	1.00	NaN	0.86	0.75
CHAID	0.84	0.94	0.75	0.90	0.87
CR-tree	0.89	0.96	0.85	0.92	0.90
Logistic Reg	0.89	0.96	0.85	0.92	0.90

**Fig. 4.** Importance variables based on the greedy model.**Table 12**

The results of the developed models using a training dataset based on the Whale model.

	Accuracy	Precision	Recall	FOR	F1_score
Quest	1.00	1.00	1.00	1.00	1.00
CHAID	0.89	0.92	0.94	0.80	0.93
CR-tree	1.00	1.00	1.00	1.00	1.00
Logistic	0.90	0.92	0.95	0.83	0.94
SVM	0.89	0.89	0.97	0.87	0.93
Neural network	0.93	0.94	0.97	0.90	0.96

Table 13

The results of the developed models using a testing dataset based on the Whale model.

	Accuracy	Precision	Recall	FOR	F1_score
Quest	0.91	0.93	0.96	0.88	0.94
CHAID	0.89	0.92	0.92	0.78	0.92
CR-tree	0.89	0.89	0.96	0.86	0.93
Logistic_Reg	0.91	1.00	0.88	0.75	0.94
SVM	0.89	0.92	0.92	0.78	0.92
Neural network	0.91	0.93	0.96	0.88	0.94

The train results are shown in Tables 14 and 15 and reveal that all prediction models' accuracy, precision, recall, FOR, and F1 score ranged between 0.7 and 1, as shown in Table 14. The results show that all the developed models can predict Parkinson's cases. The CR-tree and CHAID models performed best in the training and testing datasets, as shown in Tables 14 and 15.

Fig. 6 illustrates the results of the analysis of the importance of the PD variables. The most important variable for all the models was Speed 1, and the least important variables were DFA, RPDE, and MDVP: Shimmer.

4.7. The Parkinson's prediction model based on the proposed model

To evaluate the proposed optimizer on the Parkinson dataset, we processed the collected data using the hybrid optimizer, which combines Whale Optimization and Hybrid Grey Wolf Optimizer (GWO). The Whale Optimization model generated an initial solution, which was then passed on to the hybrid GWO for further refinement. The proposed model utilized the output of the first optimizer, which generated 15 variables, to enhance the model's accuracy, enabling a more precise classification of Parkinson's cases. Afterward, the hybrid GWO utilized the 15 variables to generate new solutions, further enhancing the prediction models. The resulting solution comprised 11 variables that exhibited superior performance compared to the initial solution.

Tables 16 and 17 show the outcomes of the prediction models constructed based on the proposed approach and assessed using the training and testing datasets. Table 16 shows that the models performed exceptionally well on the training dataset, with the Quest and CHAID models demonstrating perfect accuracy, precision, recall, FOR, and F1 scores. Though slightly less precise, the CR-tree model also performed well, especially concerning its recall and F1 score. However, the performance of the neural network, SVM, and logistic regression models varied, with the latter two showing lower accuracy and F1_score values.

Table 17 shows the performance of the trained classification models using the test dataset. Quest, CHAID, and CR-tree_status performed well, especially regarding precision and recall. The Logistic Reg and neural network models also performed well but showed slight decreases in accuracy and F1 score compared to the training dataset. Meanwhile, the SVM model experienced a notable decline in accuracy, precision, and F1 score on the testing dataset. Moreover, when the training and testing datasets were combined, the logistic regression model exhibited an overfitting problem. Consequently, the results of the logistic regression were excluded. Finally, the CR tree was selected as the best prediction model compared with others to mitigate under fitting and overfitting issues in the collected datasets.

The analysis reveals that the optimal variables vary for each predictive model. The Speed 1 variable notably influenced neural network, CRT, Quest, logistic regression, and CHAID, while the SVM model excelled in MDVP: Fo(Hz), as shown in Fig. 7. Therefore, the proposed models, incorporating important variables and their significance levels, underscore the model-specific nature of optimal variables. This highlights the importance of meticulous consideration when selecting variables for predictive modeling in future applications of Parkinson's disease or similar diseases.

4.8. Optimization algorithm summary and discussion

This section examines the optimization algorithms employed in the research. The primary objective is to identify the most crucial features for constructing a robust prediction model. Additionally, each tree model reproduces the input feature, eliminating the least significant variables from the analysis. The analysis highlights the key aspects of each model used in the research.

- **Ranker model:** The tree models eliminated numerous relatively unimportant feature variables, resulting in model variations. PPE and spread1 consistently emerged as significant variables across multiple models. Additionally, the CR-tree model exhibited exceptional accuracy (approximately 96 %) in predicting PD cases. Furthermore, assessments of F1 and precision showed the CR-tree's reliability in accurately classifying Parkinson's disease.
- **BestFirst model:** This approach was employed to construct a robust feature subset by selecting only three variables (speed1, speed2, DFA). Compared to the Ranker method, this model exhibited a more focused approach to constructing a predictive model. The proposed BestFirst models presented a simplified model that effectively

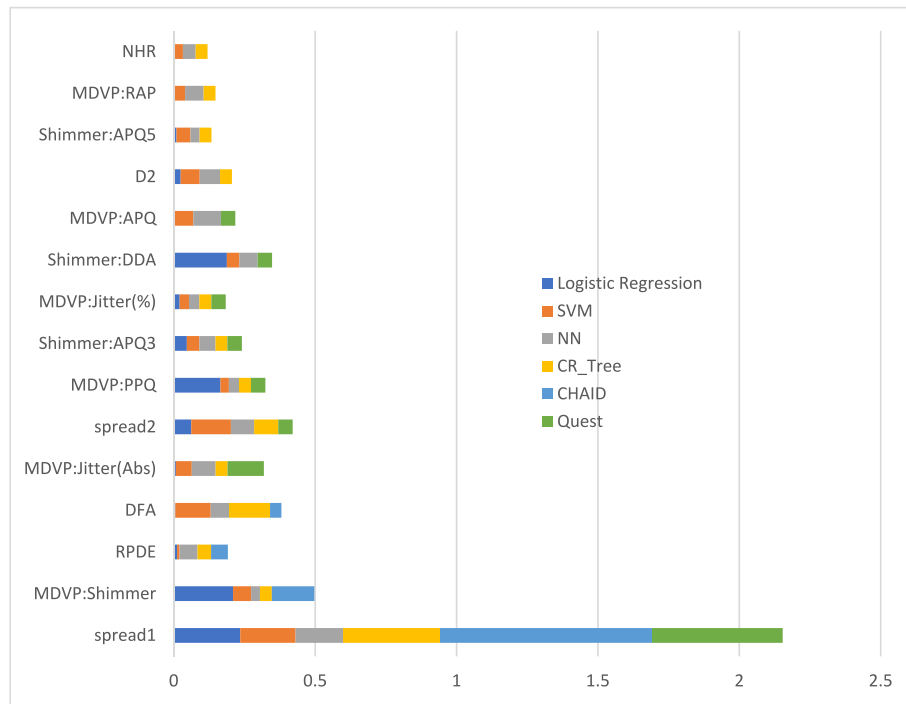


Fig. 5. Importance variables based on the Whale model.

Table 14

The results of the developed models using a training dataset based on Hybrid Grey Wolf optimization.

	Accuracy	Precision	Recall	FOR	F1_score
Quest	0.86	0.89	0.94	0.77	0.91
CHAID	0.91	0.91	0.98	0.92	0.94
CR-tree	0.88	0.92	0.92	0.75	0.92
Logistic	0.86	0.89	0.94	0.77	0.91
SVM	0.82	0.85	0.94	0.71	0.89
Neural network	0.98	0.99	0.99	0.97	0.99

The neural network achieved the best results in both datasets, with accuracy, recall, precision, FOR, and F1_score values of 0.98, 0.99, 0.99, 0.97, and 0.99, respectively.

Table 15

The results of the developed models using a testing dataset based on Hybrid Grey Wolf optimization.

	Accuracy	Precision	Recall	FOR	F1_score
Quest	0.91	0.90	0.98	0.92	0.94
CHAID	0.84	0.85	0.96	0.80	0.90
CR-tree	0.88	0.93	0.90	0.72	0.91
Logistic_Reg	0.88	0.92	0.92	0.75	0.92
SVM	0.89	0.89	0.98	0.91	0.93
Neural network	0.92	0.92	0.98	0.92	0.95

captured Parkinson's disease cases by highlighting the direct impacts of the selected features.

- **Greedy models:** The greedy model selected five important variables from the dataset, namely speed1, speed2, DFA, MDVP: RAP, and SHIMMER: APQ3, which are directly related to Parkinson's cases. All the utilized models succeeded in developing a PD prediction, except the Quest model. The greedy model was less efficient than the Ranker and BestFirst models, and it failed to develop a robust prediction model.
- **Whale Optimization:** The whale optimization selected 15 important variables from the dataset, and the selected features were used to

build PD prediction models. The main testing dataset showed that the neural network was the best model for predicting the future dataset. The whale optimization's output was examined by comparing it with previous models. The results showed that Whale optimization behaves similarly to Ranker and BestFirst, with minor changes in the selected variables. Additionally, the output of the whale model showed better results than the output of the greedy model.

- **Hybrid GWO Model:** The proposed model selected 13 variables from the dataset, showcasing its efficacy in predicting Parkinson's disease. The results of the hybrid GWO model demonstrated competitive performance compared to previous models. The neural network consistently achieved higher accuracy, precision, recall, and F1 score values than other models. Additionally, Speed 1 consistently emerged as the most important variable across all models. Conversely, DFA, RPDE, and MDVP: Shimmer were identified as the least important variables across the proposed prediction models. Finally, the testing results for all the models showed that the neural network with the hybrid GWO model was the best-performing model.
- **Proposed Optimization Model (Whale and Hybrid Grey Wolf):** The proposed model was intended to combine the nature of the whale and grey wolf optimization models to mimic the hunting behavior of humpback whales and combine strategies used by grey wolves to hunt and escape prey. The results showed that the CR-tree performed better than all previous models, with its accuracy reaching around 95 % for accuracy and exceeding 94 % for all performance metrics. The results obtained from all models showed that the proposed hybrid approach might be adept at identifying informative feature combinations that enhance the performance of the various prediction models.

4.9. Comparison between the developed models

The selected models were compared with the proposed model by employing three metrics—weighted mean [38], coefficient of dispersion (CD) [39], and coefficient of variation [40]—as illustrated in Table 18.

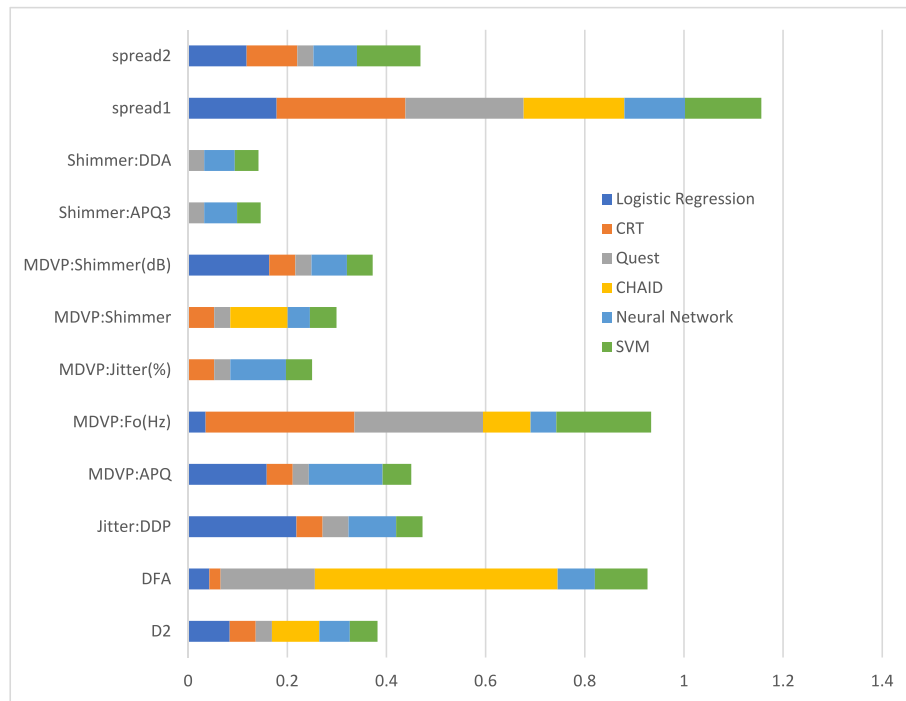


Fig. 6. Importance variables based on the hybrid GWO model.

Table 16

The results of the developed models using a training dataset based on the proposed model.

	Accuracy	Precision	Recall	FOR	F1_score
Quest	1.00	1.00	1.00	1.00	1.00
CHAID	1.00	1.00	1.00	1.00	1.00
CR-tree	0.97	0.97	0.99	0.97	0.98
Logistic regression	0.86	0.90	0.92	0.73	0.91
SVM	0.82	0.83	0.97	0.80	0.89
Neural network	1.00	1.00	1.00	1.00	1.00

Table 17

The results of the developed models using a testing dataset based on the proposed model.

	Accuracy	Precision	Recall	FOR	F1_score
Quest	0.95	0.96	0.98	0.93	0.97
CHAID	0.95	0.94	1.00	1.00	0.97
CR-tree	0.95	0.94	1.00	1.00	0.97
Logistic regression	0.92	0.94	0.96	0.87	0.95
SVM	0.81	0.80	1.00	1.00	0.89
Neural network	0.94	0.96	0.96	0.88	0.96

The findings show that the tree models outperformed the other models in the field, notably the Quest and CR-tree models. Furthermore, the neural network model outperformed the other models when utilizing the hybrid GWO model.

The analysis of the developed models revealed low variability in performance measures across models, as indicated by the coefficient of variation (CV) and coefficient of dispersion (CD) analysis. This consistency underscores reliable and stable performance across various assessment parameters. Conversely, a notable degree of variability in the false omission rate (FOR) suggests that different models differ in their ability to minimize false negatives. Across most metrics, the suggested model received the best results. Notably, it achieved a high recall and F1 score, as it efficiently identified true positives and preserved the equilibrium between recall and precision. This model could work

exceptionally well when relevant instances are critical.

Finally, the results showed that the models consistently demonstrated high performance across various metrics. The proposed model stands out with the highest scores in most metrics, achieving a perfect recall and a high F1 score. Meanwhile, variability across models was generally low, indicating stable performance.

5. Conclusion

This paper described the development and testing of a predictor of PD using biomedical sound measurements. We selected the most important factors that affect Parkinson's disease from the input dataset by using five optimization models, namely Ranker, greedy, BestFirst, hybrid GWO, and Whale optimization, to develop a Parkinson's prediction model. Afterward, the last two optimizers were merged to generate improved optimizers to solve each optimizer's weaknesses. The Parkinson's prediction models included SVM, neural network, Quest, CHAID, CR-tree, and logistic regression.

The proposed methodology started by applying one of the feature selection models, and then the model's output was forwarded to six prediction models using 70 % of the dataset. Afterward, the trained prediction models were used to validate the developed models' abilities to predict Parkinson's cases using the other 30 % of the dataset. The outputs of the six models were used as inputs for the six prediction models after being split into train and test datasets. The results indicated that the tree models performed better than other models, especially the CR-tree and Quest models. Moreover, when the hybrid GWO model was applied, the neural network model outperformed the other models.

The main findings of this study can be summarized as follows: (a) biomedical sound measurements are efficiently used to predict Parkinson's cases; (b) feature selection can improve the development of an efficient Parkinson's prediction model; (c) the variables of speed 1, speed 2, and DFA can be used with different prediction models to develop a less efficient and less complex model; (d) using all input variables to develop a predictor of Parkinson's disease is not necessary to develop an accurate and reliable model; (e) the proposed model stands out as having the highest scores in most metrics, notably

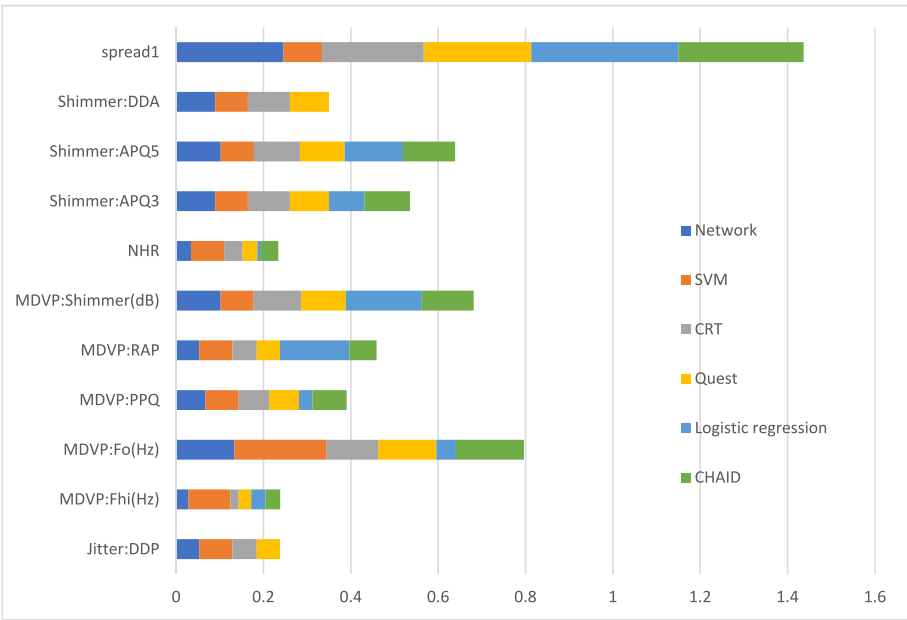


Fig. 7. Importance variables based on the proposed model.

Table 18
Ratio statistics for the developed models' performance metrics.

Metric	Ranker	BestFirst	Greedy	Whale	Hybrid GWO	Proposed	Weighted Mean	CD	CV
Accuracy	0.91	0.91	0.89	0.91	0.92	0.95	0.92	0.02	0.021
Precision	0.9	0.9	0.9	0.93	0.92	0.94	0.92	0.014	0.015
Recall	0.98	0.98	0.96	0.96	0.98	1	0.97	0.02	0.021
FOR	0.92	0.92	0.85	0.88	0.92	1	0.91	0.026	0.028
F1_score	0.94	0.94	0.92	0.94	0.95	0.97	0.94	0.015	0.016
Optimal	Quest	Quest	CR-tree	NN	NN	CR-tree	–	–	–

achieving a perfect recall and a high F1 score; and (g) variability across models was generally low, indicating stable performance.

Additionally, the number of selected samples is small, and an additional dataset is needed to improve the prediction model's performance. The proposed hybrid optimization model is complex, as it employs two optimization models to enhance the performance of the selection process. However, this complexity increases the overall complexity of the optimization model. In future work, other clinical and meta-patient variables directly affecting Parkinson's disease will be considered to develop a fast, efficient, and low-complexity prediction model. Moreover, other prediction models can be used to improve the Parkinson's prediction model with fewer errors. In the future, researchers are encouraged to use other optimization algorithms, such as practical swarm optimization, moth flame lion, ant lion optimizer, and the dragonfly algorithm, with advanced prediction models to enhance Parkinson's disease prediction. The prediction models could also include advanced deep neural network models.

Funding statement

No funds have been received for this research.

Ethical Statement

- 1) This material is the authors' own original work, which has not been previously published elsewhere.
- 2) The paper is not currently being considered for publication elsewhere.

- 3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
- 4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
- 5) The results are appropriately placed in the context of prior and existing research.
- 6) All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving proper reference.
- 7) All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content.

I agree with the above statements and declare that this submission follows the policies of Solid State Ionics as outlined in the Guide for Authors and in the Ethical Statement.

CRedit authorship contribution statement

Hazem Al-Najjar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Nadia Al-Rousan:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Dana Al-Najjar:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declared no conflict of interest.

Acknowledgment

All authors equally contributed to this research.

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